TIME SERIES ANALYSIS

EXERCISES 1 **100 Marks**

1. The activities in the forecasting process are depicted in ***Figure 1***. You are asked to provide sales forecasts of several products for a large biscuit manufacturing company. Describe each of the forecasting activity in the context of this project. **(15)**

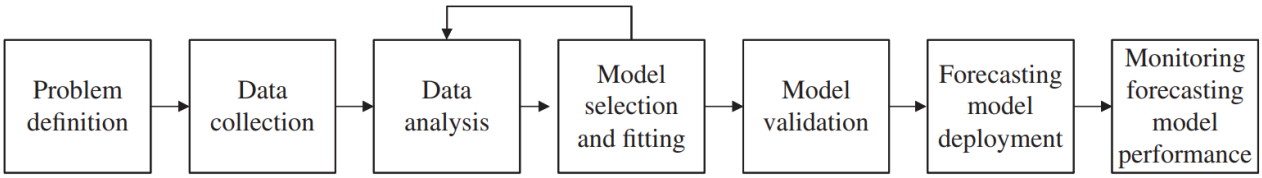


Figure 1: The forecasting process.

2. For each of the following series, what sort of time patterns would you expect to see?

   a) Monthly sales of computer hard drives for the past 10 years at your local store. **(2)**
   b) Hourly pulse rate of a person for one week. **(2)**
   c) Daily sales at a fast-food store for six months. **(2)**
   d) Weekly electricity consumption for your local area over the past 10 years. **(2)**

3. The graphs in ***Figure 2*** show plots of four time series along with their ACFs. Which ACF goes with which time series? Be sure to provide reasons why**.** **(12)**

4. The data in ***Table 1*** reflect the sales of electric knives for the period January 1991 through April 1992:

   a) What will the forecasts be for May 1992 using a 3-, 5-,7-, 9-, and 11- month moving average? **(10)**
   b) Assuming that the past pattern will continue into the future, what *k* value(s) should management select in order to minimise the errors? **(5)**

5. The data in ***Table 2*** concern the monthly sales figures of a shop which opened in January 1987 and sells gifts, souvenirs, and novelties. The shop is situated on the wharf at a beach resort town. The sales volume varies with the seasonal population of tourists. There is a large influx of visitors to the town at Christmas and for the local surfing festival, held every March since 1988. Over time, the shop has expanded its premises, range of products, and staff.

   a) Produce a time plot of the data and describe the patterns in the graph. Identify any unusual or unexpected fluctuations in the time series. **(4)**
   b) Explain why it is necessary to take logarithms of these data before fitting a model. **(2)**
   c) Calculate total sales each year and find the logarithms of these sales. Plot these against the year and fit a linear regression line to these data. **(6)**
   d) Use your regression model to predict the logarithm of the total annual sales for 1994, 1995, and 1996. Produce prediction intervals for each of your forecasts. **(4)**
   e) Transform your predictions and intervals to obtain predictions and intervals for the raw data. **(4)**

6. A classic example of a non-stationary series is the daily closing IBM stock prices. ***Figure 3*** shows the analysis of n = 369 daily closing prices for IBM stock. Explain how each plot shows the series is non-stationary and should be differenced. **(4)**

7. Consider ***Table 3*** which gives the number of strikes in the United States from 1951–1980.

   a) By studying appropriate graphs of the series, explain why an ARIMA(0,1,1) model seems appropriate. **(3)**
   b) Write this model in terms of the backshift operator. **(2)**
   c) Fit the model using R and examine the residuals. Is the model satisfactory? **(5)**
   d) Forecast three times ahead by hand. Check your forecasts with forecasts generated using R. **(4)**
   e) Create a plot of the series with forecasts and prediction intervals for the next three periods shown. **(2)**

8. ***Figure 4*** shows the annual bituminous coal production from 1920 to 1968. You decide to fit the following model to the series:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3} + \phi_4 Y_{t-4} + e_t$$

where Y_t is the coal production in year t and e_t is a white noise series.

   a) What sort of ARIMA model is this (i.e., what are *p, d*,and *q*)? **(3)**
   b) Explain why this model was chosen. **(1)**
   c) The last five values of the series are given below:

t (year)	1964	1965	1966	1967	1968
Y_t (Million tons)	467	512	534	552	545

The estimated parameters are:

c = 146.1, ϕ_1 = 0.891, ϕ_2 = −0.257, ϕ_3 = 0.392 and ϕ_4 =−0.333.

Give forecasts for the next three years (1969–1971). **(6)**

Answer ALL questions.

- Complete the relevant parts of Q4, Q5 & Q7 in R.

Table 1: Sales of electric knives for the period January 1991 through April 1992.

Time	Jan-91	Feb-91	Mar-91	Apr-91	May-91	Jun-91	Jul-91	Aug-91	Sep-91	Oct-91	Nov-91	Dec-91	Jan-92	Feb-92	Mar-92	Apr-92
Sales of electric knives	19	15	39	102	90	29	90	46	30	66	80	89	82	17	26	29

Table 2: Monthly sales figures for a souvenir shop.

	1987	**1988**	**1989**	**1990**	**1991**	**1992**	**1993**
Jan	1664.81	2499.81	4717.02	5921.1	4826.64	7615.03	10243.24
Feb	2397.53	5198.24	5702.63	5814.58	6470.23	9849.69	11266.88
Mar	2840.71	7225.14	9957.58	12421.25	9638.77	14558.4	21826.84
Apr	3547.29	4806.03	5304.78	6369.77	8821.17	11587.33	17357.33
May	3752.96	5900.88	6492.43	7609.12	8722.37	9332.56	15997.79
Jun	3714.74	4951.34	6630.8	7224.75	10209.48	13082.09	18601.53
Jul	4349.61	6179.12	7349.62	8121.22	11276.55	16732.78	26155.15
Aug	3566.34	4752.15	8176.62	7979.25	12552.22	19888.61	28586.52
Sep	5021.82	5496.43	8573.17	8093.06	11637.39	23933.38	30505.41
Oct	6423.48	5835.1	9690.5	8476.7	13606.89	25391.35	30821.33
Nov	7600.6	12600.08	15151.84	17914.66	21822.11	36024.8	46634.38
Dec	19756.21	28541.72	34061.01	30114.41	45060.69	80721.71	104660.67

Table 3: Number of strikes in the United States from 1951–1980

Year	**# of strikes**
1951	4737
1952	5117
1953	5091
1954	3468
1955	4320
1956	3825
1957	3673
1958	3694
1959	3708
1960	3333
1961	3367
1962	3614
1963	3362
1964	3655
1965	3963

Year	**# of strikes**
1966	4405
1967	4595
1968	5045
1969	5700
1970	5716
1971	5138
1972	5010
1973	5353
1974	6074
1975	5031
1976	5648
1977	5506
1978	4230
1979	4827
1980	3885

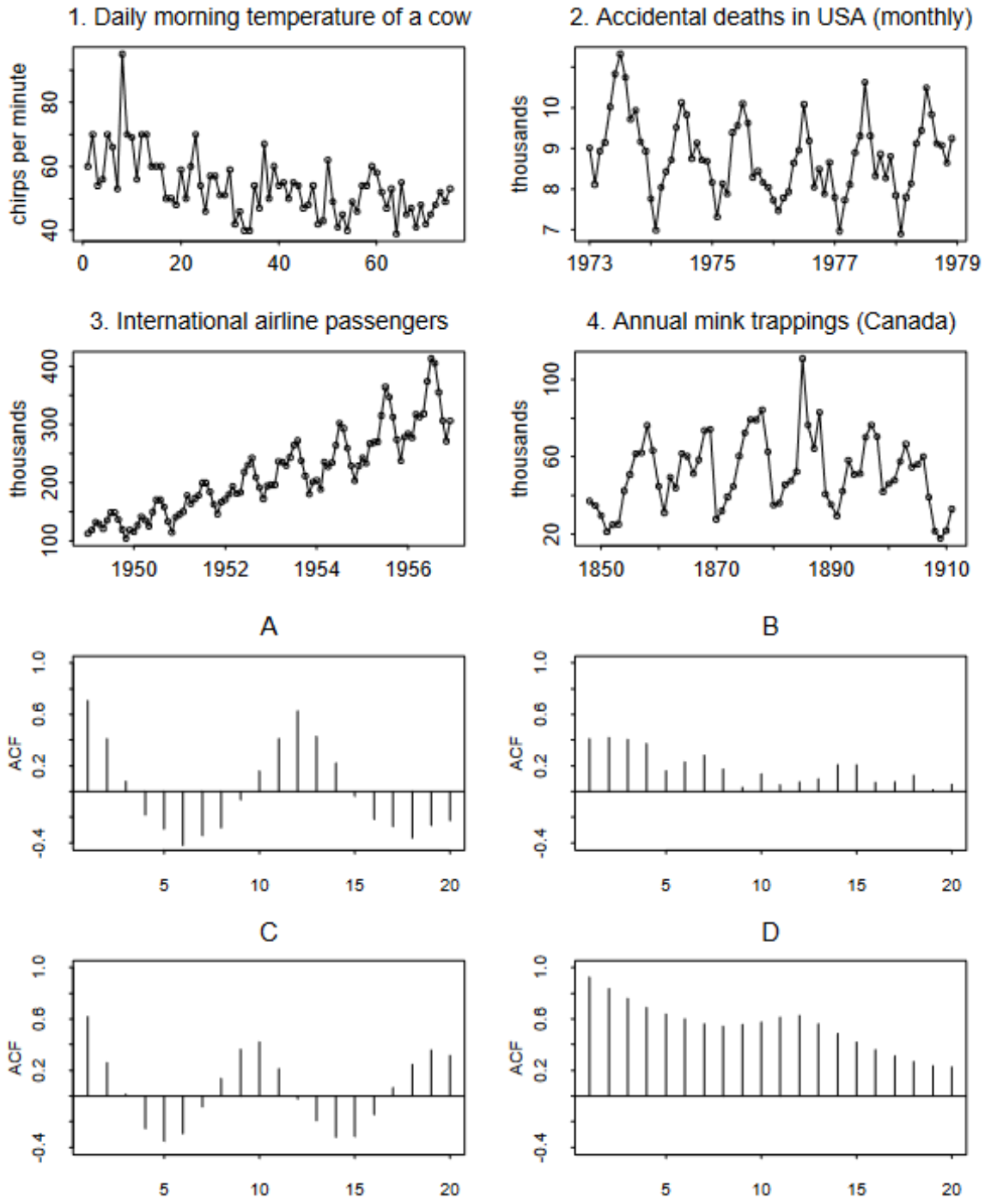


Figure 2: Four time series with their ACFs.

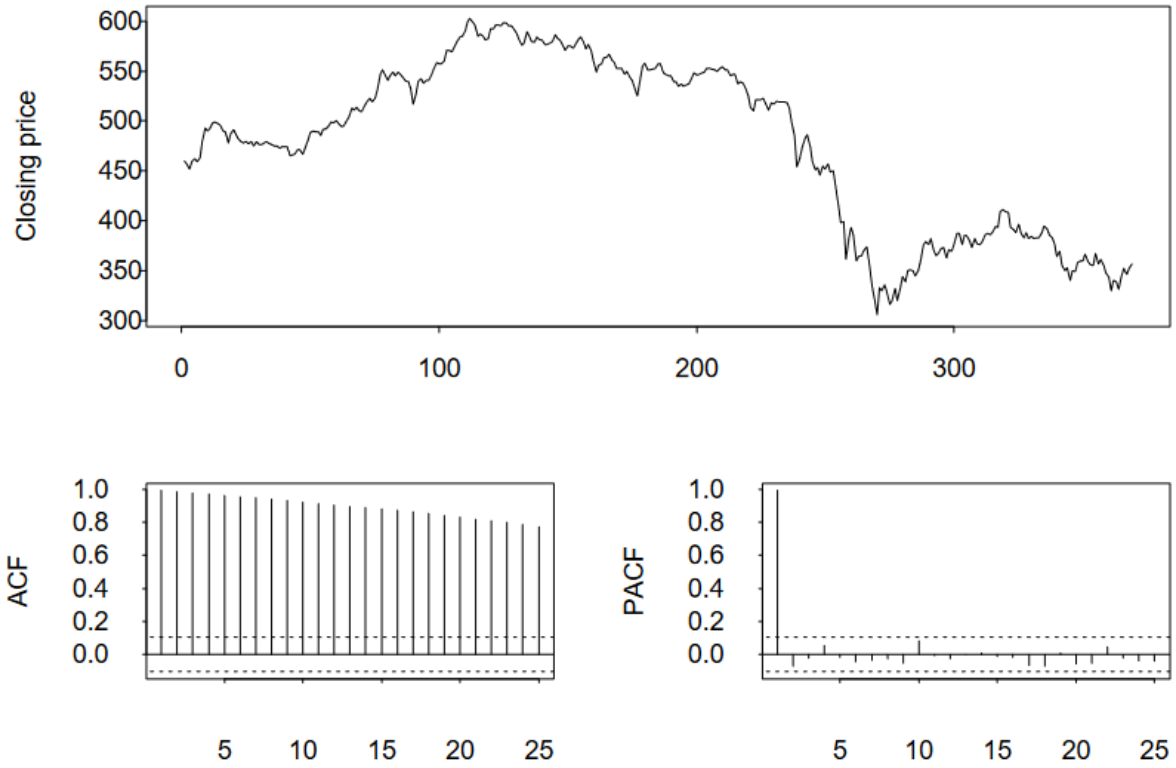


Figure 3: Time series analysis of n = 369 daily closing IBM stock prices.

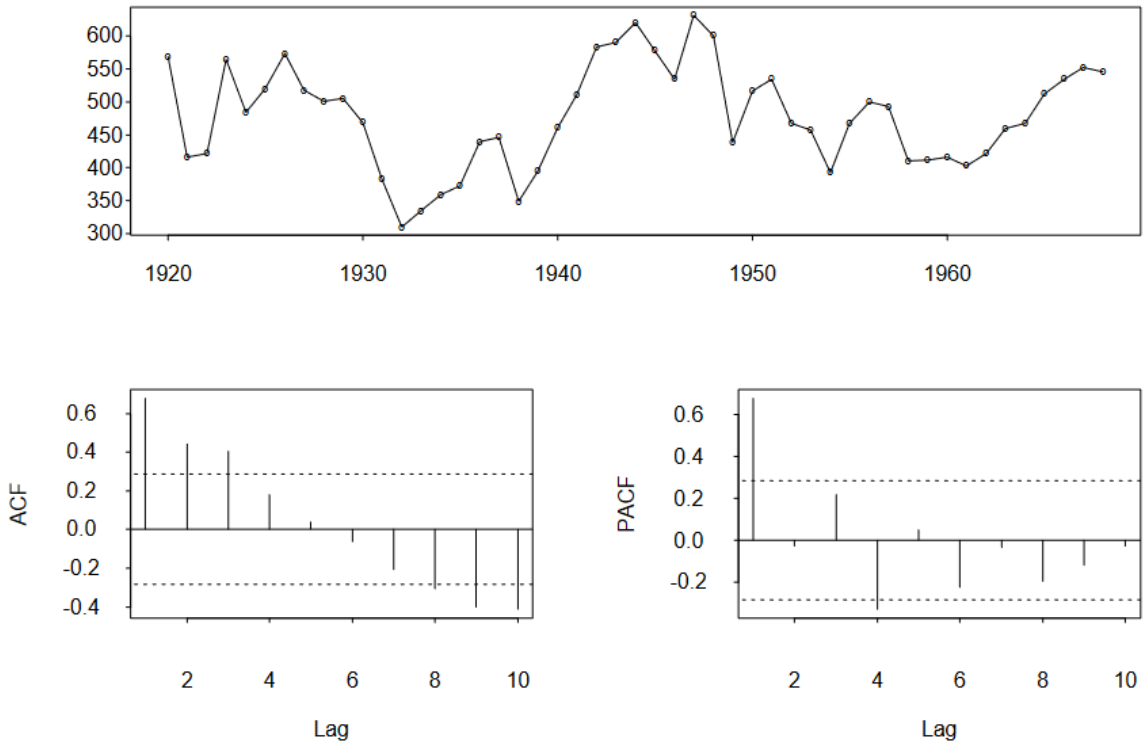


Figure 4: Annual bituminous coal production in the United States from 1920–1968.